MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
49WF POTABLE WATER TREATMENT PLANT
DESIGN OF FDN OF SC OF TANK 49WF-004 & 005

Document Number: MD-216-49WF-EG-CV-CAL-2008
Revision: D01
Revision Date: 27.10.2023
Status: AS BUILT

Outotec Document Number:	900707	GAU01	AG32	CED01	53010	OU100696991	2
L&T Document Number:	O19045	080	C	K	24398	001	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	17.09.2020	FIRST ISSUE	BJB	SJM	AG	
C01	28.12.2020	ISSUED FOR CONSTRUCTION	BJB	SJM	AG	
D01	27.10.2023	AS BUILT	BJB	SJM	AG	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1. INTENT 4

2. EXECUTIVE SUMMARY 4

3. SCOPE 4

3. DESIGN REQUIREMENTS

3.1 Design Loads 4

3.2 Load Combination 5

3.3 Stability Against Overturning & Sliding 6

4. CONCRETE FOUNDATIONS

4.1 Design 6

4.2 Materials 6

4.3 Foundations for Stair 6

4.4 Grouting 6

4.5 Minimum Concrete Cover 7

4.6 Minimum Reinforcement Steel (Rebar) Size 7

4.7 Allowable Bearing Capacity 7

5. DESIGN CALCULATION FOR FOUNDATION 8

References 25

ATTACHMENT 1 STRUCTURAL STAIR DRAWING 26

ATTACHMENT 2 STRUCTURAL STAIR STAAD OUTPUT 28

ATTACHMENT 3 F1 STAAD OUTPUT 48

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
FDN	:	Foundation
SC	:	Staircase

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This document is to Prepare for design of **Potable Water Treatment Plant (49WF)-** Stair's Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah – Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel & STAAD. For member strength calculation force values are directly taken from STAAD Output.

3. DESIGN REQUIREMENTS

3.1 Design Load

All Design Loads are directly taken from STAAD analysis of stair structure (Ref Dwg. No. MD-216-49WF-EG-ME-GAD-2004) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3.2 Load Combination

Load combinations are based on SBC-301,2007 & SBC-304,2018.

Basic Load cases are –

1. Dead Load (D)
2. Live Load (L)
3. Wind Load (WLX)(+/-)
4. Wind Load (WLZ)(+/-)

Allowable Stress Design

Load Case	Load Combination
201	D
202	D + L
203	D + 0.75 L + 0.75 WLX
204	D+0.75L+ 0.75WLXZ
205	DL + WLX
206	DL + WLZ
207	0.6 D + WLX
208	0.6 D + WLZ

Ultimate Strength Design

Load Case	Load Combination
100	1.4 (D)
101	1.2(D) + 1.6 L
102	1.2(D) + 0.8WLX
103	1.2 (D) + 0.5L+ 1.6WLX
104	0.9 (D) + 1.6WLX
105	1.2 (D) + 0.8 WLZ
106	1.2 (D) + 0.5 L+ 1.6 WLZ
107	0.9 (D) + 1.6WLZ

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3.3 Stability Against Overturning & Sliding

Overturning

Structures and building foundations, shall have the following minimum stability ratios against overturning:

- 1.5 for load combinations with wind loads
- 2 for other load combinations

Sliding

A minimum factor of safety against sliding shall be 1.5.

4. CONCRETE FOUNDATIONS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundations for Stair

Foundation shall be combined type. The design calculations as determined by the standard in Civil, Structural & Architectural design criteria (Doc no. MD-216-1000-EG-ST-CRT-1001).

4.4 Grouting

Grout shall be 30 mm thick cementitious under steel column bases of strength 40 Mpa.

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.5 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Column	50

4.6 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.7 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 650 kN/m^2 at 1.5m depth below EGL (ie. RL 917.00m) refer Table 7 - BH-28 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5. DESIGN CALCULATION FOR FOUNDATION

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Size of Columns	Along-X	Along-Z	ex	ez
Column A	0.60m	0.60m	0.0000m	-1.20m
Column B	0.60m	0.60m	0.0000m	1.20m
Size of fdn. raft:	2.40m	3.30m		
Thickness of raft:	0.50 m			
Original Gr. Level:	0.00 m			
U/S of B/Pl Above GL:	0.57 m			
U/S level of fdn:	-1.50 m			
Unit weight of conc:	25 kN/m^3			
Unit weight of soil:	18 kN/m^3			

Calculation of Weight of Foundation :

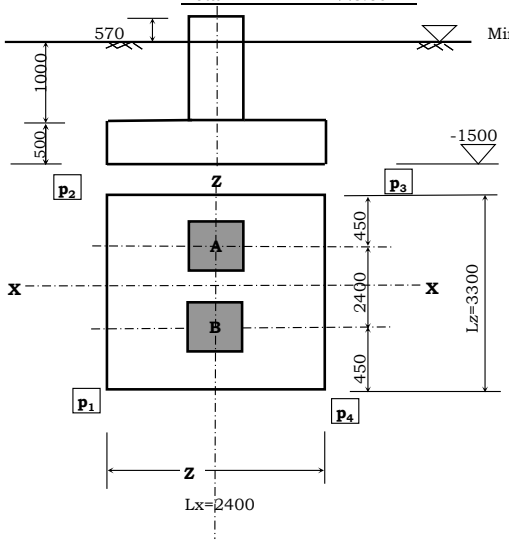
Weight of raft = (2.4*3.3*0.5*25) = 99.00KN

Weight of soil = ((2.4*3.3)-2*(0.6*0.6))*1*18 = 129.60KN

Total = 228.60KN

Design Input:

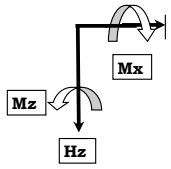
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%



Effective depth of slab along X direction = 436mm = 0.436m

Effective depth of slab along Z direction = 444mm = 0.444m

Clear projection along X direction = 0.9m



H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Basic Load Cases :

		Column A					Column B				
Sl. No	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kN-m)	Mx (kN-m)	V (kN)	Hx (kN)	Hz (kN)	Mz (kN-m)	Mx (kN-m)
1	STAIR Dead Load (D)	17	1.0	0.0	0.0	0.0	16.0	0.6	0.0	0.0	0.0
2	LIVE Load (L)	25.0	0.3	0.3	0.0	0.0	20.0	-0.3	0.0	0.0	0.0
4	Wind Load (WLX)(+/-)	-15.0	-9.0	0.0	0.0	0.0	-15.0	-9.0	0.0	0.0	0.0
5	Wind Load (WLZ)(+/-)	-93.0	0.0	-43.0	0.0	0.0	93.0	0.0	-0.4	0.0	0.0

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	D	33.0	1.6	0.0	0.0	-1.2	261.60	1.60	3.31	0.00	-1.20
202	D + L	42.0	1.6	0.0	0.3	-7.2	270.60	1.60	3.31	0.25	-6.68
203	D + O.75L + 0.75WLX	44.3	-11.9	0.0	0.2	-5.7	272.85	-11.90	-24.63	0.19	-5.31
204	D+0.75L+ 0.75WLXZ	66.8	1.6	0.0	-32.4	161.7	295.35	1.60	3.31	-32.36	94.71
205	DL+WLX	3.0	-16.4	0.0	0.0	-1.2	231.60	-16.40	-33.95	0.00	-1.20
206	DL+WLZ	33.0	1.6	0.0	-43.4	222.0	261.60	1.60	3.31	-43.40	132.16
207	0.6 D + WLX	-10.2	-17.0	0.0	0.0	-0.7	218.40	-17.04	-35.27	0.00	-0.72
208	0.6D+WLZ	19.8	1.0	0.0	-43.4	222.5	248.40	0.96	1.99	-43.40	132.64

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Ultimate Conditions

100	1.4 (D)	46.2	2.2	0.0	0.0	-1.7	366.24	2.24	4.64	0.00	-1.68
101	1.2(D) + 1.6 L	111.6	1.9	0.0	0.4	-11.0	385.92	1.92	3.97	0.40	-10.21
102	1.2(D) + 0.8WLX	15.6	-12.5	0.0	0.0	-1.4	289.92	-12.48	-25.83	0.00	-1.44
103	1.2 (D) + 0.5L+ 1.6WLX	14.1	-26.9	0.0	0.0	-4.4	288.42	-26.88	-55.64	0.00	-4.44
104	0.9 (D) + 1.6WLX	-18.3	-27.4	0.0	0.0	-1.1	187.44	-27.36	-56.64	0.00	-1.08
105	1.2(D) + 0.8WLZ	39.6	1.9	0.0	-34.7	177.1	313.92	1.92	3.97	-34.72	105.25
106	1.2 (D) + 0.5L+ 1.6WLZ	62.1	1.9	0.0	0.0	352.7	336.42	1.92	3.97	0.00	352.68
107	0.9 (D) + 1.6WLZ	29.7	1.4	0.0	-69.4	356.0	235.44	1.44	2.98	-69.44	212.30

Check for overturning and sliding:

Load Cases:		Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
		V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201	D	261.60	1.60	3.31	0.00	1.20	94.78	359.70	65.40	
202	D + L	270.60	1.60	3.31	0.25	6.68	98.04	66.81	67.65	389.66
203	D + O.75L + 0.75WLX	272.85	-11.90	-24.63	0.19	5.31	13.29	84.75	9.17	523.87
204	D+0.75L+ 0.75WLXZ	295.35	1.60	3.31	-32.36	94.71	107.01	5.15	73.84	3.29
205	DL+WLX	231.60	-16.40	-33.95	0.00	1.20	8.19	318.45	5.65	
206	DL+WLZ	261.60	1.60	3.31	-43.40	132.16	94.78	3.27	65.40	2.17
207	0.6 D + WLX	218.40	-17.04	-35.27	0.00	0.72	7.43	500.50	5.13	
208	0.6D+WLZ	248.40	0.96	1.99	-43.40	132.64	150.00	3.09	103.50	2.06

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

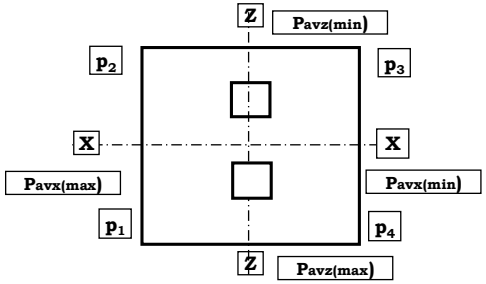
Max. Base pressure = p_{max} = (V/(Lx.Lz)) + (6.My/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)

Min. Base pressure = p_{min} = (V/(Lx.Lz)) - (6.My/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)

Surcharge load = 20 kN/m^2

Net safe bearing capacity of soil for DL, LL= 650 kN/m^2 For Wind load/Seis. load case 812.5 kN/m^2

Gross safe bearing capacity of soil for DL, LL= 677 kN/m^2 For Wind load/Seis. load case 839.5 kN/m^2



Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												Remarks
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	
201	D	261.60	3.31	-1.20	33.03	1.05	0.28	34.35	33.80	32.26	31.71	7.35	34.08	31.99	33.31	32.76	**O.K**
202	D + L	270.60	3.31	-6.68	34.17	1.05	1.53	36.75	33.68	34.66	31.59	9.75	35.22	33.13	35.71	32.64	**O.K**
203	D + O.75L + 0.75WLX	272.85	-24.63	-5.31	34.45	7.78	1.22	43.45	41.01	27.89	25.46	16.45	42.23	26.68	35.67	33.24	**O.K**
204	D+0.75L+ 0.75WLXZ	295.35	3.31	94.71	37.29	1.05	21.74	60.08	16.59	57.99	14.50	33.08	38.34	36.25	59.04	15.55	**O.K**
205	DL+WLX	231.60	-33.95	-1.20	29.24	10.72	0.28	40.23	39.68	18.80	18.25	13.23	39.96	18.53	29.52	28.97	**O.K**
206	DL+WLZ	261.60	3.31	132.16	33.03	1.05	30.34	64.42	3.74	62.33	1.64	37.42	34.08	31.99	63.38	2.69	**O.K**
207	0.6 D + WLX	218.40	-35.27	-0.72	27.58	11.13	0.17	38.88	38.54	16.61	16.28	11.88	38.71	16.45	27.75	27.41	**O.K**
208	0.6D+WLZ	248.40	1.99	132.64	31.36	0.63	30.45	62.44	1.54	61.19	0.29	35.44	31.99	30.74	61.82	0.92	**O.K**
100	1.4 (D)	366.24	4.64	-1.68	46.24	1.46	0.39	48.09	47.32	45.16	44.39	21.09	47.71	44.78	46.63	45.86	
101	1.2(D) + 1.6 L	385.92	3.97	-10.21	48.73	1.25	2.34	52.33	47.64	49.82	45.13	25.33	49.99	47.48	51.08	46.39	
102	1.2(D) + 0.8WLX	289.92	-25.83	-1.44	36.61	8.15	0.33	45.09	44.43	28.78	28.12	18.09	44.76	28.45	36.94	36.28	
103	1.2 (D) + 0.5L+ 1.6WLX	288.42	-55.64	-4.44	36.42	17.56	1.02	55.00	52.96	19.87	17.83	28.00	53.98	18.85	37.44	35.40	
104	0.9 (D) + 1.6WLX	187.44	-56.64	-1.08	23.67	17.88	0.25	41.79	41.30	6.04	5.54	14.79	41.55	5.79	23.92	23.42	
105	1.2(D) + 0.8WLZ	313.92	3.97	105.25	39.64	1.25	24.16	65.05	16.73	62.54	14.22	38.05	40.89	38.38	63.80	15.48	
106	1.2 (D) + 0.5L+ 1.6WLZ	336.42	3.97	352.68	42.48	1.25	80.96	124.70	-37.23	122.19	-39.74	97.70	43.74	41.23	123.45	-38.49	
107	0.9 (D) + 1.6WLZ	235.44	2.98	212.30	29.73	0.94	48.74	79.41	-18.07	77.52	-19.95	52.41	30.67	28.79	78.47	-19.01	

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.6+2*0.5*0.444)+(0.6+2*0.5*0.444))*2

= 4.176 m

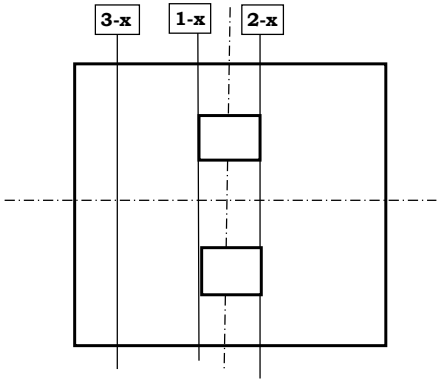
Punching Area= ((0.6+2*0.5*0.444)*(0.6+2*0.5*0.444))

= 1.09 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.5*25+1*18)

= 30.50 kN/m^2

Area of footing, A_f= 7.92 m^2



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Load Case	Load Combination	p_{1x}	p_{2x}	p_{3x}	$B.M_{1x}$	$B.M_{2x}$	$S.F_{3x}$	$S.F_{(punching)}$
100	1.4 (D)	46.61	45.88	47.14	6.82	5.93	7.85	107.52
101	1.2(D) + 1.6 L	49.05	48.42	49.50	7.77	7.00	8.93	124.49
102	1.2(D) + 0.8WLX	38.64	34.57	41.61	4.95	0.00	5.89	41.70
103	1.2 (D) + 0.5L+ 1.6WLX	40.81	32.02	47.19	7.73	-2.94	9.32	40.41
104	0.9 (D) + 1.6WLX	28.14	19.20	34.64	2.66	-8.20	3.52	-46.67
105	1.2(D) + 0.8WLZ	39.95	39.32	40.40	4.08	3.32	4.71	62.40
106	1.2 (D) + 0.5L+ 1.6WLZ	42.80	42.17	43.25	5.24	4.47	6.03	81.80
107	0.9 (D) + 1.6WLZ	29.97	29.50	30.31	-0.03	-0.60	-0.01	-5.28

Overhang = 0.9 m
Effective depth = 0.436 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	7.77	0.04	0.05	0.000	56.60	784.80	16ϕT	256.20	300	200	1005.31
At top	-8.20	0.04	0.06	0.000	0.00	392.40	12ϕT	288.22	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 9.32 kN/m

Shear stress , τ_v= (9.32*1000)/(1000*436)

= 0.02 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 124.49 kN

Strength of Concrete in shear Vc1 = 5385.81 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 3669.64 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 3590.54 kN **Table 22.6.5.2_ (c)**

Design Vc= 3590.54 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (124.49*1000)/(4176*444)

= 0.07 N/mm^2 **No shear reinforcement reqd.**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Calculation of distribution reinf in Z direction :

Effective depth = 0.444 m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	27.25	0.14	0.18	0.000	195.44	799.20	16ϕT	251.58	300	200	1005.31
At top	23.54	0.12	0.16	0.000	168.78	399.60	12ϕT	283.03	300	200	565.49

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 45.38 kN/m

Shear stress , τ_v= (45.38*1000)/(1000*444)

= 0.10 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**12ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**12ϕT**	**200**	c/c

PROPRIETARY & CONFIDENTIAL

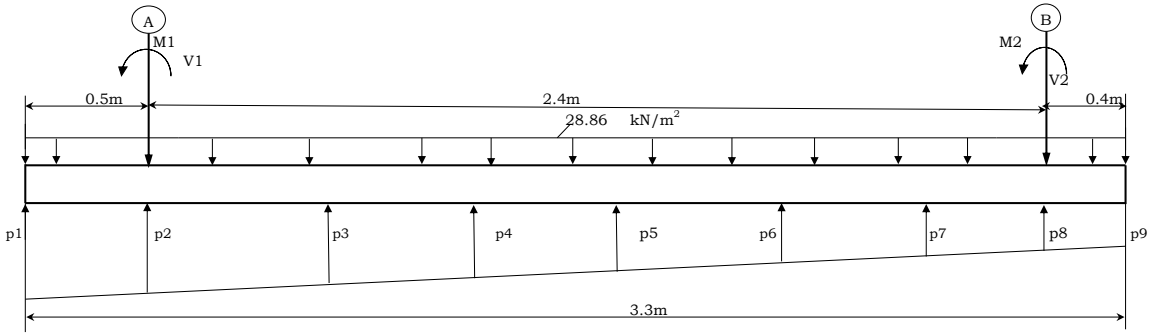
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

CALCULATION OF BM & SF :

(Along Z direction)



Load Case	Load Combination	Soil pressure in UDL (kN/m)								
		p1	p2	p3	p4	p5	p6	p7	p8	p9
100	1.4 (D)	114.49								107.46
101	1.2(D) + 1.6 L	119.96								113.94
102	1.2(D) + 0.8WLX	107.42								68.28
103	1.2 (D) + 0.5L+ 1.6WLX	129.55			By STAAD					45.24
104	0.9 (D) + 1.6WLX	99.71								13.90
105	1.2(D) + 0.8WLZ	98.14								92.11
106	1.2 (D) + 0.5L+ 1.6WLZ	104.96								98.94
107	0.9 (D) + 1.6WLZ	73.61								69.08

From STAAD Output	
BM at top	56.50 (kNm/m)
BM at bot.	65.40 (kNm/m)
SF (T. at top)	106.30 (kN/m)
SF (T. at bottom)	108.90 (kN/m)

Load Case	Load Combination	Vertical Load (kN/m)							Moment (kNm/m)						
		V1	V2	V3	V4	V5	V6	V7	M1	M2	M3	M4	M5	M6	M7
100	1.4 (D)	23.80			22.40				4.64						
101	1.2(D) + 1.6 L	60.40			51.20				3.97						
102	1.2(D) + 0.8WLX	8.40			7.20				-25.83						
103	1.2 (D) + 0.5L+ 1.6WLX	8.90			5.20				-55.64						
104	0.9 (D) + 1.6WLX	-8.70			-9.60				-56.64						
105	1.2(D) + 0.8WLZ	-54.00			93.6				3.97						
106	1.2 (D) + 0.5L+ 1.6WLZ	-115.90			178				3.97						
107	0.9 (D) + 1.6WLZ	-133.50			163.2				2.98						

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Beam Force Detail

Sign convention as diagrams:- positive above line, negative below line except Fx where positive is compression. Distance d is given from beam end A.

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
1	100:1.4 (D)	0.000	0.000	-0.000	0.000	0.000	0.000	0.000
		0.045	0.000	3.851	0.000	0.000	0.000	-0.096
		0.090	0.000	7.698	0.000	0.000	0.000	-0.361
		0.135	0.000	11.540	0.000	0.000	0.000	-0.794
		0.180	0.000	15.379	0.000	0.000	0.000	-1.395
		0.225	0.000	19.213	0.000	0.000	0.000	-2.163
		0.270	0.000	23.042	0.000	0.000	0.000	-3.124
		0.315	0.000	26.867	0.000	0.000	0.000	-4.252
		0.360	0.000	30.688	0.000	0.000	0.000	-5.547
		0.405	0.000	34.505	0.000	0.000	0.000	-7.009
		0.450	-0.000	38.318	-0.000	-0.000	-0.000	-8.638
	101:1.2(D) +	0.000	0.000	-0.000	0.000	0.000	0.000	-0.000
		0.045	0.000	4.097	0.000	0.000	0.000	-0.102
		0.090	0.000	8.191	0.000	0.000	0.000	-0.384
		0.135	0.000	12.282	0.000	0.000	0.000	-0.845
		0.180	0.000	16.368	0.000	0.000	0.000	-1.484
		0.225	0.000	20.451	0.000	0.000	0.000	-2.303
		0.270	0.000	24.530	0.000	0.000	0.000	-3.325
		0.315	0.000	28.606	0.000	0.000	0.000	-4.525
		0.360	0.000	32.678	0.000	0.000	0.000	-5.904
		0.405	0.000	36.746	0.000	0.000	0.000	-7.461
		0.450	-0.000	40.811	-0.000	-0.000	-0.000	-9.196
	102:1.2(D) + (	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
		0.045	0.000	3.522	0.000	0.000	0.000	-0.088
		0.090	0.000	7.020	0.000	0.000	0.000	-0.330
		0.135	0.000	10.496	0.000	0.000	0.000	-0.724
		0.180	0.000	13.948	0.000	0.000	0.000	-1.270
		0.225	0.000	17.376	0.000	0.000	0.000	-1.966
		0.270	0.000	20.778	0.000	0.000	0.000	-2.833
		0.315	0.000	24.157	0.000	0.000	0.000	-3.848
		0.360	0.000	27.512	0.000	0.000	0.000	-5.011
		0.405	0.000	30.844	0.000	0.000	0.000	-6.320
		0.450	-0.000	34.153	-0.000	-0.000	-0.000	-7.774
	103:1.2(D) +	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
		0.045	0.000	4.502	0.000	0.000	0.000	-0.113
		0.090	0.000	8.954	0.000	0.000	0.000	-0.421
		0.135	0.000	13.356	0.000	0.000	0.000	-0.923
		0.180	0.000	17.707	0.000	0.000	0.000	-1.617
		0.225	0.000	22.008	0.000	0.000	0.000	-2.500
		0.270	0.000	26.252	0.000	0.000	0.000	-3.597
		0.315	0.000	30.445	0.000	0.000	0.000	-4.878
		0.360	0.000	34.588	0.000	0.000	0.000	-6.341
		0.405	0.000	38.681	0.000	0.000	0.000	-7.985
		0.450	-0.000	42.723	-0.000	-0.000	-0.000	-9.807
	104:0.9 (D) +	0.000	0.000	-0.000	0.000	0.000	0.000	0.000
		0.045	0.000	3.159	0.000	0.000	0.000	-0.079

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		0.090	0.000	6.267	0.000	0.000	0.000	-0.295
		0.135	0.000	9.323	0.000	0.000	0.000	-0.646
		0.180	0.000	12.329	0.000	0.000	0.000	-1.130
		0.225	0.000	15.283	0.000	0.000	0.000	-1.744
		0.270	0.000	18.179	0.000	0.000	0.000	-2.504
		0.315	0.000	21.023	0.000	0.000	0.000	-3.390
		0.360	0.000	23.817	0.000	0.000	0.000	-4.399
		0.405	0.000	26.559	0.000	0.000	0.000	-5.530
		0.450	-0.000	29.250	-0.000	-0.000	-0.000	-6.779
	105:1.2(D) + (	0.000	0.000	0.000	0.000	0.000	0.000	0.000
		0.045	0.000	3.116	0.000	0.000	0.000	-0.078
		0.090	0.000	6.227	0.000	0.000	0.000	-0.292
		0.135	0.000	9.336	0.000	0.000	0.000	-0.642
		0.180	0.000	12.440	0.000	0.000	0.000	-1.128
		0.225	0.000	15.541	0.000	0.000	0.000	-1.750
		0.270	0.000	18.638	0.000	0.000	0.000	-2.527
		0.315	0.000	21.731	0.000	0.000	0.000	-3.439
		0.360	0.000	24.821	0.000	0.000	0.000	-4.487
		0.405	0.000	27.907	0.000	0.000	0.000	-5.669
		0.450	-0.000	30.989	-0.000	-0.000	-0.000	-6.987
	106:1.2(D) +	0.000	0.000	-0.000	0.000	0.000	0.000	0.000
		0.045	0.000	3.422	0.000	0.000	0.000	-0.086
		0.090	0.000	6.841	0.000	0.000	0.000	-0.321
		0.135	0.000	10.257	0.000	0.000	0.000	-0.706
		0.180	0.000	13.668	0.000	0.000	0.000	-1.240
		0.225	0.000	17.076	0.000	0.000	0.000	-1.923
		0.270	0.000	20.480	0.000	0.000	0.000	-2.776
		0.315	0.000	23.881	0.000	0.000	0.000	-3.779
		0.360	0.000	27.278	0.000	0.000	0.000	-4.930
		0.405	0.000	30.671	0.000	0.000	0.000	-6.229
		0.450	-0.000	34.061	-0.000	-0.000	-0.000	-7.677
	107:0.9 (D) +	0.000	0.000	-0.000	0.000	0.000	0.000	-0.000
		0.045	0.000	2.012	0.000	0.000	0.000	-0.050
		0.090	0.000	4.022	0.000	0.000	0.000	-0.189
		0.135	0.000	6.028	0.000	0.000	0.000	-0.415
		0.180	0.000	8.033	0.000	0.000	0.000	-0.729
		0.225	0.000	10.034	0.000	0.000	0.000	-1.130
		0.270	0.000	12.032	0.000	0.000	0.000	-1.632
		0.315	0.000	14.028	0.000	0.000	0.000	-2.220
		0.360	0.000	16.020	0.000	0.000	0.000	-2.897
		0.405	0.000	18.011	0.000	0.000	0.000	-3.660
		0.450	-0.000	19.998	-0.000	-0.000	-0.000	-4.510
2	100:1.4 (D)	0.000	0.000	-97.896	0.000	0.000	0.000	-3.998
		0.240	0.000	-77.644	0.000	0.000	0.000	16.795
		0.480	0.000	-57.510	0.000	0.000	0.000	32.876
		0.720	0.000	-37.496	0.000	0.000	0.000	44.274
		0.960	0.000	-17.601	0.000	0.000	0.000	51.016
		1.200	0.000	2.175	0.000	0.000	0.000	53.129

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		1.440	0.000	21.814	0.000	0.000	0.000	49.986
		1.680	0.000	41.334	0.000	0.000	0.000	42.275
		1.920	0.000	60.735	0.000	0.000	0.000	30.024
		2.160	0.000	80.017	0.000	0.000	0.000	13.260
		2.400	-0.000	99.180	-0.000	-0.000	-0.000	-7.991
	101:1.2(D) +	0.000	0.000	-105.161	0.000	0.000	0.000	-5.226
		0.240	0.000	-83.552	0.000	0.000	0.000	17.129
		0.480	0.000	-62.045	0.000	0.000	0.000	34.455
		0.720	0.000	-40.641	0.000	0.000	0.000	46.775
		0.960	0.000	-19.339	0.000	0.000	0.000	54.113
		1.200	0.000	1.861	0.000	0.000	0.000	56.491
		1.440	0.000	22.944	0.000	0.000	0.000	53.231
		1.680	0.000	43.925	0.000	0.000	0.000	45.064
		1.920	0.000	64.804	0.000	0.000	0.000	32.015
		2.160	0.000	85.581	0.000	0.000	0.000	14.105
		2.400	-0.000	106.255	-0.000	-0.000	-0.000	-8.642
	102:1.2(D) + (	0.000	0.000	-88.752	0.000	0.000	0.000	-33.604
		0.240	0.000	-71.556	0.000	0.000	0.000	-14.609
		0.480	0.000	-55.025	0.000	0.000	0.000	0.456
		0.720	0.000	-39.158	0.000	0.000	0.000	11.742
		0.960	0.000	-23.955	0.000	0.000	0.000	19.403
		1.200	0.000	-9.417	0.000	0.000	0.000	23.588
		1.440	0.000	4.363	0.000	0.000	0.000	23.999
		1.680	0.000	17.477	0.000	0.000	0.000	21.276
		1.920	0.000	29.928	0.000	0.000	0.000	15.572
		2.160	0.000	41.714	0.000	0.000	0.000	7.039
		2.400	-0.000	52.836	-0.000	-0.000	-0.000	-4.172
	103:1.2(D) +	0.000	0.000	-108.921	0.000	0.000	0.000	-65.447
		0.240	0.000	-88.333	0.000	0.000	0.000	-42.078
		0.480	0.000	-69.175	0.000	0.000	0.000	-23.336
		0.720	0.000	-51.448	0.000	0.000	0.000	-8.894
		0.960	0.000	-35.152	0.000	0.000	0.000	1.575
		1.200	0.000	-20.286	0.000	0.000	0.000	8.399
		1.440	0.000	-7.055	0.000	0.000	0.000	11.476
		1.680	0.000	4.746	0.000	0.000	0.000	11.643
		1.920	0.000	15.115	0.000	0.000	0.000	9.227
		2.160	0.000	24.055	0.000	0.000	0.000	4.555
		2.400	-0.000	31.563	-0.000	-0.000	-0.000	-2.047
	104:0.9 (D) +	0.000	0.000	-72.907	0.000	0.000	0.000	-63.419
		0.240	0.000	-59.543	0.000	0.000	0.000	-47.731
		0.480	0.000	-47.636	0.000	0.000	0.000	-34.980
		0.720	0.000	-37.184	0.000	0.000	0.000	-24.835
		0.960	0.000	-28.189	0.000	0.000	0.000	-16.962
		1.200	0.000	-20.650	0.000	0.000	0.000	-11.029
		1.440	0.000	-14.775	0.000	0.000	0.000	-6.884
		1.680	0.000	-10.357	0.000	0.000	0.000	-3.930
		1.920	0.000	-7.395	0.000	0.000	0.000	-1.833
		2.160	0.000	-5.889	0.000	0.000	0.000	-0.260

Beam Force Detail Cont...

			Axial	Shear		Torsion	Bending	
Beam	L/C	d (m)	Fx (kN)	Fy (kN)	Fz (kN)	Mx (kN-m)	My (kN-m)	Mz (kN-m)
		2.400	-0.000	-5.839	-0.000	-0.000	-0.000	1.120
	105:1.2(D) + (	0.000	0.000	-78.965	0.000	0.000	0.000	-3.017
		0.240	0.000	-62.595	0.000	0.000	0.000	13.750
		0.480	0.000	-46.328	0.000	0.000	0.000	26.711
		0.720	0.000	-30.163	0.000	0.000	0.000	35.887
		0.960	0.000	-14.100	0.000	0.000	0.000	41.304
		1.200	0.000	1.861	0.000	0.000	0.000	42.983
		1.440	0.000	17.705	0.000	0.000	0.000	40.422
		1.680	0.000	33.447	0.000	0.000	0.000	34.177
		1.920	0.000	49.086	0.000	0.000	0.000	24.270
		2.160	0.000	64.624	0.000	0.000	0.000	10.726
		2.400	-0.000	80.059	-0.000	-0.000	-0.000	-6.432
	106:1.2(D) +	0.000	0.000	-87.161	0.000	0.000	0.000	-3.707
		0.240	0.000	-69.152	0.000	0.000	0.000	14.808
		0.480	0.000	-51.245	0.000	0.000	0.000	29.134
		0.720	0.000	-33.441	0.000	0.000	0.000	39.294
		0.960	0.000	-15.739	0.000	0.000	0.000	45.311
		1.200	0.000	1.861	0.000	0.000	0.000	47.209
		1.440	0.000	19.344	0.000	0.000	0.000	44.430
		1.680	0.000	36.725	0.000	0.000	0.000	37.583
		1.920	0.000	54.004	0.000	0.000	0.000	26.693
		2.160	0.000	71.181	0.000	0.000	0.000	11.783
		2.400	-0.000	88.255	-0.000	-0.000	-0.000	-7.123
	107:0.9 (D) +	0.000	0.000	-50.572	0.000	0.000	0.000	-1.530
		0.240	0.000	-40.025	0.000	0.000	0.000	9.199
		0.480	0.000	-29.554	0.000	0.000	0.000	17.477
		0.720	0.000	-19.160	0.000	0.000	0.000	23.321
		0.960	0.000	-8.843	0.000	0.000	0.000	26.749
		1.200	0.000	1.397	0.000	0.000	0.000	27.778
		1.440	0.000	11.550	0.000	0.000	0.000	26.087
		1.680	0.000	21.625	0.000	0.000	0.000	22.037
		1.920	0.000	31.624	0.000	0.000	0.000	15.646
		2.160	0.000	41.547	0.000	0.000	0.000	6.930
		2.400	-0.000	51.392	-0.000	-0.000	-0.000	-4.093
3	100:1.4 (D)	0.000	0.000	-35.586	0.000	0.000	0.000	-7.991
		0.045	0.000	-32.008	0.000	0.000	0.000	-6.479
		0.090	0.000	-28.435	0.000	0.000	0.000	-5.123
		0.135	0.000	-24.865	0.000	0.000	0.000	-3.924
		0.180	0.000	-21.300	0.000	0.000	0.000	-2.881
		0.225	0.000	-17.739	0.000	0.000	0.000	-1.994
		0.270	0.000	-14.183	0.000	0.000	0.000	-1.284
		0.315	0.000	-10.631	0.000	0.000	0.000	-0.730
		0.360	0.000	-7.083	0.000	0.000	0.000	-0.332
		0.405	0.000	-3.539	0.000	0.000	0.000	-0.088
		0.450	-0.000	0.000	-0.000	-0.000	-0.000	-0.000
	101:1.2(D) +	0.000	0.000	-38.471	0.000	0.000	0.000	-8.642
		0.045	0.000	-34.607	0.000	0.000	0.000	-7.007
		0.090	0.000	-30.747	0.000	0.000	0.000	-5.542

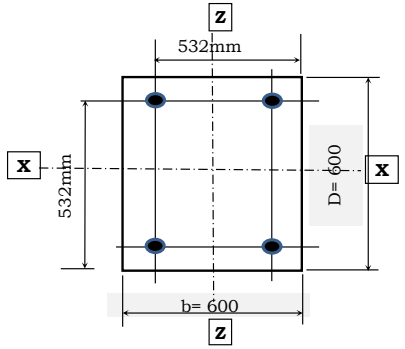
Beam Force Detail Cont...

Beam	L/C	d	Axial	Shear		Torsion	Bending	
			Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.135	0.000	-26.891	0.000	0.000	0.000	-4.245
		0.180	0.000	-23.038	0.000	0.000	0.000	-3.117
		0.225	0.000	-19.189	0.000	0.000	0.000	-2.157
		0.270	0.000	-15.344	0.000	0.000	0.000	-1.390
		0.315	0.000	-11.503	0.000	0.000	0.000	-0.790
		0.360	0.000	-7.665	0.000	0.000	0.000	-0.359
		0.405	0.000	-3.831	0.000	0.000	0.000	-0.096
		0.450	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	102:1.2(D) + (	0.000	0.000	-18.941	0.000	0.000	0.000	-4.172
		0.045	0.000	-16.940	0.000	0.000	0.000	-3.369
		0.090	0.000	-14.962	0.000	0.000	0.000	-2.654
		0.135	0.000	-13.008	0.000	0.000	0.000	-2.025
		0.180	0.000	-11.077	0.000	0.000	0.000	-1.481
		0.225	0.000	-9.170	0.000	0.000	0.000	-1.020
		0.270	0.000	-7.289	0.000	0.000	0.000	-0.655
		0.315	0.000	-5.432	0.000	0.000	0.000	-0.371
		0.360	0.000	-3.598	0.000	0.000	0.000	-0.168
		0.405	0.000	-1.787	0.000	0.000	0.000	-0.045
		0.450	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	103:1.2(D) +	0.000	0.000	-9.959	0.000	0.000	0.000	-2.047
		0.045	0.000	-8.733	0.000	0.000	0.000	-1.629
		0.090	0.000	-7.557	0.000	0.000	0.000	-1.264
		0.135	0.000	-6.432	0.000	0.000	0.000	-0.950
		0.180	0.000	-5.357	0.000	0.000	0.000	-0.684
		0.225	0.000	-4.332	0.000	0.000	0.000	-0.463
		0.270	0.000	-3.365	0.000	0.000	0.000	-0.293
		0.315	0.000	-2.448	0.000	0.000	0.000	-0.163
		0.360	0.000	-1.582	0.000	0.000	0.000	-0.073
		0.405	0.000	-0.766	0.000	0.000	0.000	-0.019
		0.450	-0.000	-0.000	-0.000	-0.000	-0.000	0.000
	104:0.9 (D) +	0.000	0.000	4.100	0.000	0.000	0.000	1.120
		0.045	0.000	3.924	0.000	0.000	0.000	0.940
		0.090	0.000	3.696	0.000	0.000	0.000	0.768
		0.135	0.000	3.418	0.000	0.000	0.000	0.608
		0.180	0.000	3.089	0.000	0.000	0.000	0.461
		0.225	0.000	2.708	0.000	0.000	0.000	0.329
		0.270	0.000	2.269	0.000	0.000	0.000	0.218
		0.315	0.000	1.778	0.000	0.000	0.000	0.128
		0.360	0.000	1.237	0.000	0.000	0.000	0.060
		0.405	0.000	0.644	0.000	0.000	0.000	0.016
		0.450	-0.000	0.000	-0.000	-0.000	-0.000	0.000
	105:1.2(D) + (	0.000	0.000	-28.647	0.000	0.000	0.000	-6.432
		0.045	0.000	-25.766	0.000	0.000	0.000	-5.215
		0.090	0.000	-22.888	0.000	0.000	0.000	-4.124
		0.135	0.000	-20.014	0.000	0.000	0.000	-3.158
		0.180	0.000	-17.144	0.000	0.000	0.000	-2.319
		0.225	0.000	-14.277	0.000	0.000	0.000	-1.604
		0.270	0.000	-11.415	0.000	0.000	0.000	-1.034

Beam Force Detail Cont...

Beam	L/C	d	Axial	Shear		Torsion	Bending	
			Fx	Fy	Fz	Mx	My	Mz
		(m)	(kN)	(kN)	(kN)	(kN-m)	(kN-m)	(kN-m)
		0.315	0.000	-8.556	0.000	0.000	0.000	-0.588
		0.360	0.000	-5.700	0.000	0.000	0.000	-0.267
		0.405	0.000	-2.848	0.000	0.000	0.000	-0.071
		0.450	-0.000	0.000	-0.000	-0.000	-0.000	0.000
	106:1.2(D) +	0.000	0.000	-31.721	0.000	0.000	0.000	-7.123
		0.045	0.000	-28.532	0.000	0.000	0.000	-5.776
		0.090	0.000	-25.347	0.000	0.000	0.000	-4.567
		0.135	0.000	-22.166	0.000	0.000	0.000	-3.498
		0.180	0.000	-18.988	0.000	0.000	0.000	-2.568
		0.225	0.000	-15.814	0.000	0.000	0.000	-1.777
		0.270	0.000	-12.644	0.000	0.000	0.000	-1.145
		0.315	0.000	-9.478	0.000	0.000	0.000	-0.651
		0.360	0.000	-6.315	0.000	0.000	0.000	-0.296
		0.405	0.000	-3.156	0.000	0.000	0.000	-0.079
		0.450	-0.000	0.000	-0.000	-0.000	-0.000	0.000
	107:0.9 (D) +	0.000	0.000	-18.239	0.000	0.000	0.000	-4.093
		0.045	0.000	-16.402	0.000	0.000	0.000	-3.318
		0.090	0.000	-14.569	0.000	0.000	0.000	-2.624
		0.135	0.000	-12.738	0.000	0.000	0.000	-2.009
		0.180	0.000	-10.910	0.000	0.000	0.000	-1.475
		0.225	0.000	-9.084	0.000	0.000	0.000	-1.021
		0.270	0.000	-7.262	0.000	0.000	0.000	-0.657
		0.315	0.000	-5.442	0.000	0.000	0.000	-0.374
		0.360	0.000	-3.626	0.000	0.000	0.000	-0.170
		0.405	0.000	-1.811	0.000	0.000	0.000	-0.045
		0.450	-0.000	0.000	-0.000	-0.000	-0.000	-0.000

DESIGN OF COLUMN MKD **C1**



Input:

Clear cover to reinf.=	50mm	Table 20.6.1.3.1
Dia. of main reinf.=	16mm	
Dia. of link=	10mm	
Effective cover =	68mm	
Grade of Concrete (f_c') =	M35	
Grade of Steel (f_y) =	Fy420	
Grade of Reinf(links).=	Fy420	
Min. Steel Ratio	0.50%	
Max. Steel Ratio	4.00%	

For Compression-controlled sections

Strength reduction factor ϕ= 0.65 Table 21.2.2

Unsupproted length of column about X (l_{ux}) =	1570mm
Unsupproted length of column about Z (l_{uz}) =	1570mm

For Columns not braced against sidesway,

Where, Effective length factor K= 1.2

Slenderness ratio,	K l_{ux} / r_x =	10.88	< 22	SECTION 6.2.5a	Radius of gyration r_x=	173 mm
	K l_{uz} / r_z =	10.88	< 22	SECTION 6.2.5a	Radius of gyration r_z=	173 mm

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sl. No.	Basic Load Case	V (kN)	Hx (kN)	Hz (kN)	Mz (kNm)	Mx (kNm)
1	STAIR Dead Load (D)	17.0	1.0	0.0	0.0	0.0
2	LIVE Load (L)	25.0	0.3	0.3	0.0	0.0
3	Wind Load (WLX)(+/-)	-15.0	9.0	0.0	0.0	0.0
4	Wind Load (WLZ)(+/-)	-93.0	0.0	43.0	0.0	0.0

γ_y = 0.81

γ_z = 0.81

Load Case	Load Combination	Factored Load/Moment			Nominal Load/Moment			Kn = P_n/f_cbD (N/mm^2)
		Pu (T)	Muz (Tm)	Mux (Tm)	Pn (T)	Mnz (Tm)	Mnx (Tm)	
100	1.4 (D)	0.00	0.0	0.0	0.0	0.0	0.0	0.00
101	1.2(D) + 1.6 L	0.0	0.0	0.0	0.0	0.0	0.0	0.00
102	1.2(D) + 0.8WLX	-1.2	1.1	0.0	-1.8	1.7	0.0	0.00
103	1.2 (D) + 0.5L+ 1.6WLX	-2.4	2.3	0.0	-3.7	3.5	0.0	-0.003
104	0.9 (D) + 1.6WLX	-2.4	2.3	0.0	-3.7	3.5	0.0	-0.003
105	1.2(D) + 0.8WLZ	-7.4	0.0	5.4	-11.4	0.0	8.3	-0.01
106	1.2 (D) + 0.5L+ 1.6WLZ	-14.9	0.0	10.8	-22.9	0.0	16.6	-0.02
107	0.9 (D) + 1.6WLZ	-14.9	0.0	10.8	-22.9	0.0	16.6	-0.02

Reinforcement: Provide,

4-Nos	20 mm Dia	1257.14 Sq mm (At Corner)	A_{sc} pvd. = 3771.43 Sq mm
4-Nos	20 mm Dia	1257.14 Sq mm (Along Z axis)	
4-Nos	20 mm Dia	1257.14 Sq mm (Along Y axis)	Total A_{sc} = 1.05 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = $0.85 f'_c (A_g - A_{sc}) + f_y A_{sc}$ = 1218.18 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.8		M_{noz}	M_{nox}	β	$\alpha_n = \log(0.5)/\log(\beta)$	Interaction ratio	Remarks
	$R_{nz}=M_{noz}/f_cb^2D$	$R_{nx}=M_{nox}/f_cb^2D$						
100	0.95	0.95	718.2	718.2	0.55	1.16	0.0000	O.K
101	0.95	0.95	718.2	718.2	0.55	1.16	0.00	O.K
102	0.95	0.95	718.2	718.2	0.55	1.16	0.00	O.K
103	0.95	0.95	718.2	718.2	0.55	1.16	0.00	O.K
104	0.03	0.03	22.7	22.7	0.55	1.16	0.11	O.K
105	0.03	0.03	22.7	22.7	0.55	1.16	0.31	O.K
106	0.03	0.03	22.7	22.7	0.55	1.16	0.70	O.K
107	0.03	0.03	22.7	22.7	0.55	1.16	0.70	O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCE

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

PROPRIETARY & CONFIDENTIAL

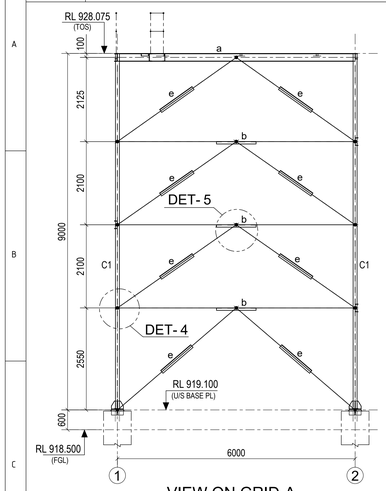
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 STRUCTURAL STAIR DRAWING

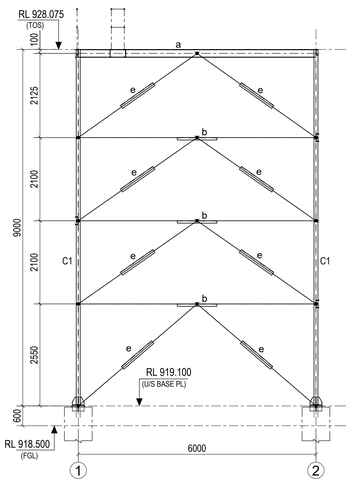
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

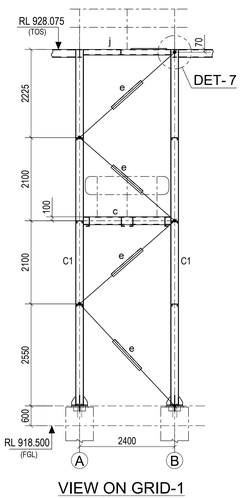
DRAWING NUMBER: MD-216-49WF-EG-ME-GAD-2004 | REV: D01



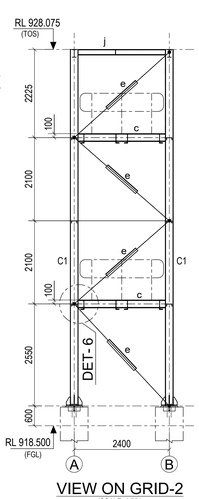
VIEW ON GRID-A
(SCALE- 1:75)



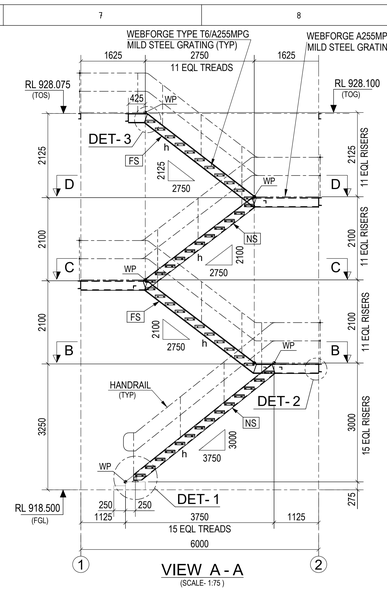
VIEW ON GRID-B
(SCALE- 1:75)



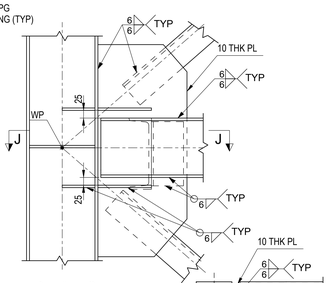
VIEW ON GRID-1
(SCALE- 1:75)

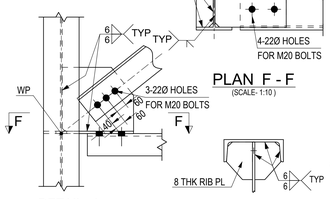


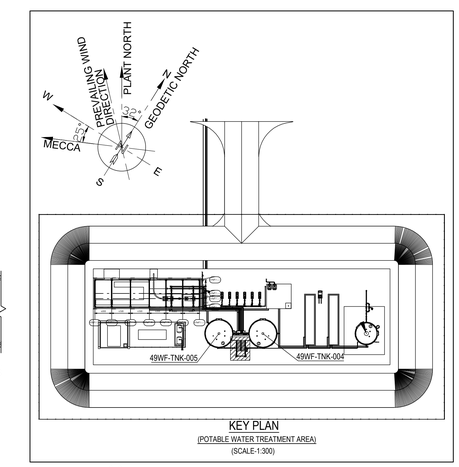
VIEW ON GRID-2
(SCALE- 1:75)

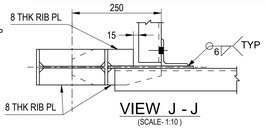


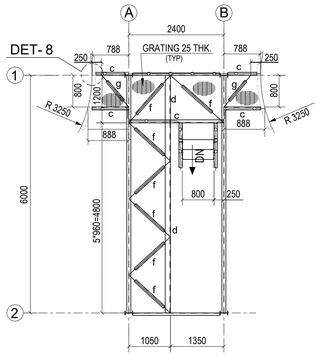
VIEW A - A
(SCALE- 1:75)

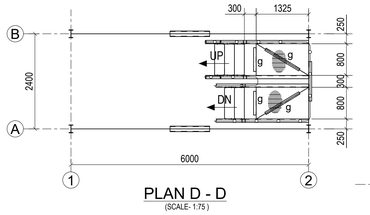


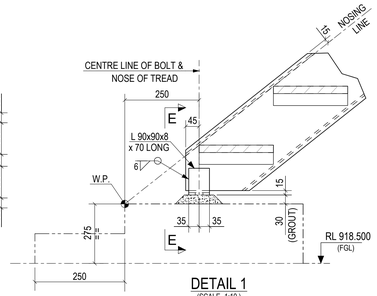


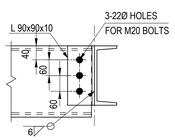


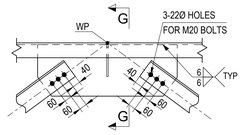


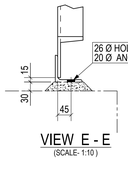


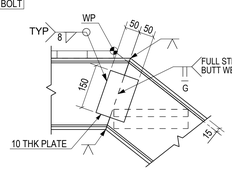


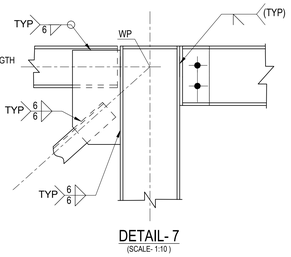


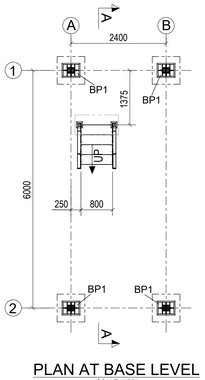


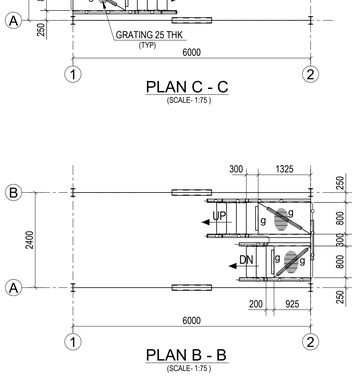


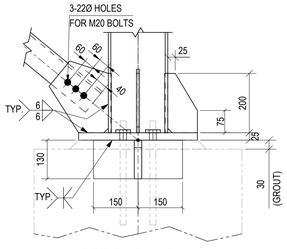


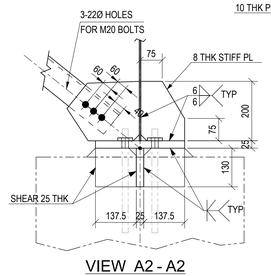


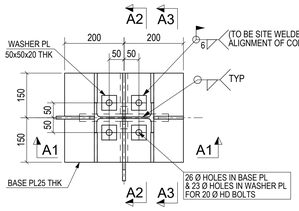


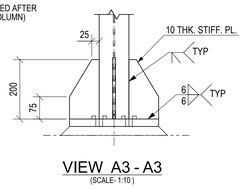


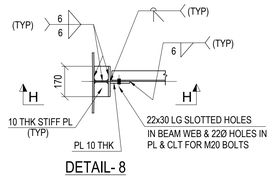


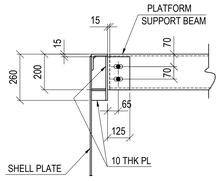












SCHEDULE OF MEMBERS

MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(T)	FACTORED SHEAR FORCE(T)	REMARKS
C1	IPE200	I	-	-	MATERIAL GRADE S275 JR
a	IPE200	I	12	-	
b	UPN200	⌐	-	-	
c	UPN200	]	-	-	
d	L - 130X130X10	⌝	-	-	
e	L - 90X90X8	⌝	12	-	
f	L - 65X65X6	⌝	-	-	
g	L - 50X50X6	⌝	-	-	
h	UPN240	]	-	-	
j	UPN140	]	-	-	

ALL GUSSET PLATE ARE 8mm THK. (UNO)
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION.
FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.

NOTES:

1. ALL DIMENSIONS ARE IN MM & LEVELS IN METRES UNLESS OTHERWISE SPECIFIED.
2. ANCHOR BOLT OF 20mm DIA. MATERIAL SHALL BE ASTM F 1554 Gr.36, CONFORMING TO THE REQUIREMENTS OF ASTM F 1554
3. ALL WELDS SHALL BE OF SHOP QUALITY AND TO BE TESTED AS PER QUALITY ASSURANCE PLAN (QAP).
4. LIGHTING PURPOSE SUPPORT DETAIL IN THE STAIR CASE AND CONNECTING PLATFORM WILL BE INDICATED IN SEPARATE DRAWING.
5. FOR ALL OTHER NOTES REFER DRAWING MD - 216 - 49WF -EG -ME - GAD - 1001.
6. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
7. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NOS MD - 216 - 49WF -EG -ME - GAD - 2001 , 2002 & 2003
8. FOR LEGEND LIST REFER DRAWING MD - 216 - 49WF -EG -ME - GAD - 2001.
9. FOR HANDRAIL & STAIR RELATED DETAILS REFER DRG. NO. MD-216-1000-EG-ST-DED-1002 & 1003 (LATEST).
10. ALL STRUCTURAL BOLTS SHALL BE AS PER ASTM A325M.

LEGENDS

TOS - TOP OF STEEL
U/S - UNDER SIDE
DN - DOWN
TYP - TYPICAL
THK - THICKNESS
PL - PLATE
WP - WORK POINT
RL - REDUCED LEVEL
UNO - UNLESS NOTED OTHERWISE
MKD - MARKED
NS - NEAR SIDE
FS - FAR SIDE

THIS DRAWING SUPERSEDES MAADEN DRAWING NUMBER MD-216-49WF-EG-ST-GAD-1004-C01 DATE: 07/03/2020

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	D01	27/10/2023	AS BUILT	SDR	SYC	KDU	PRI	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	C02	03/09/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	SDR	SYC	KDU	PRI	
MD-216-49WF-EG-ME-GAD-1008	POTABLE WATER STORAGE TANK 49WF-TNK-004	C01	07/03/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED C01	SJH	SYC	KDU	PRI	
MD-216-49WF-EG-ME-GAD-1009	POTABLE WATER STORAGE TANK 49WF-TNK-005	A01	31/01/2020	ISSUED FOR REFERENCE	SWB	SYC	KDU	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER	019045_080_S_G_26210_004	REVISION 3
OUTOTEC DRAWING NUMBER	900707_GAU01_AG32_MTL01_53062	DOCUMENT ID 0U100671398 / REVISION 3

DRAWN	SWB	29/01/2020
CHECKED	SYC	30/01/2020
ENGINEER	KDU	28/01/2020
APPROVED	PRI	31/01/2020
EMPLOYER		
SCALE	1:75,15,10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
POTABLE WATER SUPPLY, TRT & DIST
STAIRCASE OF TANK 49WF-TNK-004, 005

DRAWING NUMBER: MD-216-49WF-EG-ME-GAD-2004 | REV: D01

ATTACHMENT 2 STRUCTURAL STAIR STAAD OUTPUT

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    SEP 14, 2020                *
*           Time=     9:39:31                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\MAADEN\Potable Water Treatment Plant\STAIR\STAAD\BoosterPumpStn-StairCase.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 04-JAN-20
    4. END JOB INFORMATION
    5. INPUT WIDTH 79
    6. UNIT METER KN
    7. JOINT COORDINATES
    8. 1 0 0 0; 2 6 0 0; 3 0 0 2.4; 4 6 0 2.4; 5 0 10.175 0; 6 6 10.175 0
    9. 7 0 10.175 2.4; 8 6 10.175 2.4; 9 0 2.2 0; 10 0 2.2 2.4; 11 6 2.2 0
   10. 12 6 2.2 2.4; 13 0 4.825 0; 14 0 4.825 2.4; 15 6 4.825 0; 16 6 4.825 2.4
   11. 17 0 7.45 0; 18 0 7.45 2.4; 19 6 7.45 0; 20 6 7.45 2.4; 21 3 10.175 0
   12. 22 3 10.175 2.4; 23 3 2.2 2.4; 24 3 4.825 2.4; 25 3 7.45 2.4; 26 3 2.2 0
   13. 27 3 4.825 0; 28 3 7.45 0; 29 0 10.175 1.2; 30 6 10.175 1.2; 31 1 10.175 2.4
   14. 32 1 10.175 0; 33 1 10.175 1.2; 34 5 10.175 0; 35 5 10.175 2.4
   15. 36 5 10.175 1.2; 38 2 10.175 1.2; 39 4 10.175 1.2
   16. MEMBER INCIDENCES
   17. 1 1 9; 2 3 10; 3 2 11; 4 4 12; 5 5 32; 6 7 31; 7 5 29; 8 6 30; 9 9 13
   18. 10 10 14; 11 11 15; 12 12 16; 13 13 17; 14 14 18; 15 15 19; 16 16 20; 17 17 5
   19. 18 18 7; 19 19 6; 20 20 8; 21 10 23; 22 14 24; 23 18 25; 24 9 26; 25 13 27
   20. 26 17 28; 27 15 16; 29 9 10; 30 17 18; 31 2 12; 32 12 15; 33 15 20; 34 1 10
   21. 35 10 13; 36 13 18; 37 6 20; 39 5 18; 41 21 34; 42 22 35; 43 23 12; 44 24 16
   22. 45 25 20; 46 26 11; 47 27 15; 48 28 19; 49 3 23; 50 4 23; 51 10 24; 52 12 24
   23. 53 14 25; 54 16 25; 55 18 22; 56 20 22; 57 1 26; 58 2 26; 59 9 27; 60 11 27
   24. 61 13 28; 62 15 28; 63 17 21; 64 19 21; 65 29 7; 66 30 8; 67 31 22; 68 32 21
   25. 69 32 33; 70 29 33; 71 33 31; 72 33 38; 73 34 6; 74 35 8; 75 34 36; 76 32 29
   26. 77 29 31; 78 34 30; 79 30 35; 80 36 30; 81 36 35; 82 39 38; 86 39 36; 87 31 38
   27. 88 38 22; 89 22 39; 90 39 35
   28. SUPPORTS
   29. 1 TO 4 PINNED
   30. MEMBER RELEASE
   31. 5 TO 8 21 TO 27 29 TO 37 39 49 TO 64 69 70 72 75 TO 80 87 TO 90 START MY MZ
   32. 27 29 TO 37 39 43 TO 66 70 71 73 76 TO 81 86 TO 90 END MY MZ
   33. DEFINE MATERIAL START
   34. ISOTROPIC STEEL
   35. E 2.05E+008
   36. POISSON 0.3
   37. DENSITY 76.8195
   38. ALPHA 1.2E-005
```

```
39. DAMP 0.03
40. TYPE STEEL
41. STRENGTH FY 275200 FU 407800 RY 1.5 RT 1.2
42. END DEFINE MATERIAL
43. MEMBER PROPERTY EUROPEAN
44. 1 TO 6 9 TO 20 41 42 67 68 73 74 TABLE ST IPE200
45. 7 8 21 TO 26 43 TO 48 65 66 TABLE ST UPN140
46. 69 71 72 82 86 TABLE ST L75X75X8
47. 27 29 30 75 81 TABLE ST UPN200
48. 70 76 TO 80 87 TO 90 TABLE ST L65X65X6
49. 31 TO 37 39 49 TO 64 TABLE ST L90X90X8
50. CONSTANTS
51. BETA 90 MEMB 1 TO 4 9 TO 26 43 TO 48
52. MATERIAL STEEL ALL
53. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD
54. SELFWEIGHT Y -1
55. MEMBER LOAD
56. 27 29 30 75 CON GY -2 0.25
57. 27 29 30 75 CON GY -2 1.05
58. 27 29 30 CON GY -2 1.35
59. 27 29 30 CON GY -2 2.15
60. * LOAD FROM 800 WIDE WALKWAY
61. 73 CON GY -0.5 0.2
62. JOINT LOAD
63. 6 FY -0.5
64. MEMBER LOAD
65. 8 66 75 81 UNI GY -0.33
66. **
67. CALCULATE RAYLEIGH FREQUENCY
68. **
69. LOAD 2 LOADTYPE LIVE  TITLE LIVE LOAD
70. MEMBER LOAD
71. 27 29 30 75 CON GY -5.2 0.25
72. 27 29 30 75 CON GY -5.2 1.05
73. 27 29 30 CON GY -5.2 1.35
74. 27 29 30 CON GY -5.2 2.15
75. * LIVE LOAD FROM 800 WIDE WALKWAY
76. 73 CON GY -1.76 0.2
77. JOINT LOAD
78. 6 FY -1.76
79. MEMBER LOAD
80. 8 66 75 81 UNI GY -2
81. LOAD 3 LOADTYPE WIND  TITLE WIND LOAD +X
82. MEMBER LOAD
83. 1 TO 4 9 TO 20 UNI GX 0.86
84. LOAD 5 LOADTYPE WIND  TITLE WIND LOAD +Z
85. MEMBER LOAD
86. 1 TO 4 9 TO 20 UNI GZ 2.15
87. *******LOAD COMBINATRIONS FOR DESIGN*****************************************
88. LOAD COMB 100 1.4*DL
89. 1 1.4
90. LOAD COMB 101 1.2DL+1.6*LL
91. 1 1.2 2 1.6
92. LOAD COMB 102 1.2DL+0.8WLX
93. 1 1.2 3 0.8
94. LOAD COMB 103 1.2DL+0.5LL+1.6WLX
```

```
 95. 1 1.2 2 0.5 3 1.6
 96. LOAD COMB 104 0.9DL+1.6WLX
 97. 1 0.9 3 1.6
 98. LOAD COMB 105 1.2DL+0.8WLZ
 99. 1 1.2
100. LOAD COMB 106 1.2DL+0.5LL+1.6WLZ
101. 1 1.2 2 0.5
102. LOAD COMB 107 0.9DL+1.6WLZ
103. 1 0.9
104. ************************************************
105. *******LOAD COMBS FOR DEFLECTION***************************
106. LOAD COMB 201 DL+LL
107. 1 1.0 2 1.0
108. LOAD COMB 202 DL+LL+WLX
109. 1 1.0 2 1.0 3 1.0
110. LOAD COMB 203 DL+WLX
111. 1 1.0 3 1.0
112. LOAD COMB 204 DL+LL+WLZ
113. 1 1.0 2 1.0
114. LOAD COMB 205 DL+WLZ
115. 1 1.0
116. ************************************************************
117. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**38**	**NUMBER OF MEMBERS**	**84**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 28/ 10/ 63 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 216
TOTAL LOAD COMBINATION CASES = 13 SO FAR.
SIZE OF STIFFNESS MATRIX = 14 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.3/ 324845.0 MB

****
* *
* **RAYLEIGH FREQUENCY FOR LOADING 1 = 9.86233 CPS** *
* **MAX DEFLECTION = 0.87904 CM GLO Y, AT JOINT 38** *
* *
****

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE NONE TITLE DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.283135316E+01
Y = 0.591864674E+01
Z = 0.114866685E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -62.01
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 71.23 MY= 0.00 MZ= -175.57

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 62.01
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -71.23 MY= 0.00 MZ= 175.57

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	-2.07711E-03	17
Y =	-8.79042E-01	38
Z =	-5.10403E-03	6
RX=	-3.52216E-04	22
RY=	-9.23836E-05	7
RZ=	5.50533E-03	39

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE LIVE TITLE LIVE LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.290167602E+01
Y = 0.628952513E+01
Z = 0.108426443E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -85.92
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 93.16 MY= 0.00 MZ= -249.31

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 85.92
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -93.16 MY= 0.00 MZ= 249.31

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	4.44904E-03	31
Y =	-1.80309E-01	36
Z =	-1.89587E-02	6
RX=	-3.67472E-04	36
RY=	5.36635E-05	6
RZ=	-6.55786E-04	21

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE WIND TITLE WIND LOAD +X

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.299999999E+01
Y = 0.508749984E+01
Z = 0.119999999E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 35.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 42.00 MZ= -178.07

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -35.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -42.00 MZ= 178.07

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	6.70778E-02	5
Y =	-9.38003E-03	35
Z =	2.96667E-03	12
RX=	-3.92502E-05	32
RY=	4.53962E-04	6
RZ=	1.44000E-03	5

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE WIND TITLE WIND LOAD +Z

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.299999999E+01
Y = 0.508749979E+01
Z = 0.119999999E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 87.51

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 445.18 MY= -262.52 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -87.51

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -445.18 MY= 262.52 MZ= -0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	-3.08388E-03	31
Y =	-5.88447E-02	18
Z =	7.06007E-01	19
RX=	1.74925E-03	2
RY=	-4.28065E-04	28
RZ=	-3.06555E-05	30

********** END OF DATA FROM INTERNAL STORAGE **********

118. **
119. PRINT SUPPORT REACTION LIST 1 TO 36

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	1	0.76	17.10	0.09	0.00	0.00	0.00
	2	0.25	24.37	0.23	0.00	0.00	0.00
	3	-8.77	-14.95	-0.03	0.00	0.00	0.00
	5	-0.03	-92.95	-43.37	0.00	0.00	0.00
	100	1.06	23.94	0.13	0.00	0.00	0.00
	101	1.31	59.51	0.48	0.00	0.00	0.00
	102	-6.10	8.56	0.09	0.00	0.00	0.00
	103	-12.99	8.78	0.18	0.00	0.00	0.00
	104	-13.34	-8.54	0.04	0.00	0.00	0.00
	105	0.91	20.52	0.11	0.00	0.00	0.00
	106	1.03	32.70	0.22	0.00	0.00	0.00
	107	0.68	15.39	0.08	0.00	0.00	0.00
	201	1.01	41.47	0.32	0.00	0.00	0.00
	202	-7.76	26.52	0.29	0.00	0.00	0.00
	203	-8.01	2.15	0.06	0.00	0.00	0.00
	204	1.01	41.47	0.32	0.00	0.00	0.00
	205	0.76	17.10	0.09	0.00	0.00	0.00
2	1	-0.59	15.23	-0.01	0.00	0.00	0.00
	2	0.24	22.73	-0.09	0.00	0.00	0.00
	3	-8.77	14.95	0.04	0.00	0.00	0.00
	5	-0.03	-92.54	-42.18	0.00	0.00	0.00
	100	-0.82	21.33	-0.01	0.00	0.00	0.00
	101	-0.31	54.65	-0.16	0.00	0.00	0.00
	102	-7.72	30.24	0.02	0.00	0.00	0.00
	103	-14.61	53.57	0.01	0.00	0.00	0.00
	104	-14.56	37.63	0.06	0.00	0.00	0.00
	105	-0.70	18.28	-0.01	0.00	0.00	0.00
	106	-0.58	29.64	-0.05	0.00	0.00	0.00
	107	-0.53	13.71	-0.01	0.00	0.00	0.00
	201	-0.34	37.96	-0.10	0.00	0.00	0.00
	202	-9.11	52.92	-0.06	0.00	0.00	0.00
	203	-9.35	30.19	0.03	0.00	0.00	0.00
	204	-0.34	37.96	-0.10	0.00	0.00	0.00
	205	-0.59	15.23	-0.01	0.00	0.00	0.00
3	1	0.59	15.65	-0.02	0.00	0.00	0.00
	2	-0.25	20.00	-0.03	0.00	0.00	0.00
	3	-8.73	-14.73	0.01	0.00	0.00	0.00
	5	0.04	92.95	-0.41	0.00	0.00	0.00
	100	0.82	21.91	-0.03	0.00	0.00	0.00
	101	0.31	50.77	-0.08	0.00	0.00	0.00
	102	-6.28	7.00	-0.01	0.00	0.00	0.00
	103	-13.39	5.22	-0.02	0.00	0.00	0.00
	104	-13.44	-9.48	0.00	0.00	0.00	0.00
	105	0.70	18.78	-0.03	0.00	0.00	0.00
	106	0.58	28.78	-0.04	0.00	0.00	0.00
	107	0.53	14.08	-0.02	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	201	0.34	35.65	-0.05	0.00	0.00	0.00
	202	-8.39	20.92	-0.04	0.00	0.00	0.00
	203	-8.15	0.92	-0.01	0.00	0.00	0.00
	204	0.34	35.65	-0.05	0.00	0.00	0.00
	205	0.59	15.65	-0.02	0.00	0.00	0.00
4	1	-0.76	14.03	-0.06	0.00	0.00	0.00
	2	-0.25	18.82	-0.11	0.00	0.00	0.00
	3	-8.74	14.73	-0.03	0.00	0.00	0.00
	5	0.02	92.54	-1.55	0.00	0.00	0.00
	100	-1.06	19.64	-0.09	0.00	0.00	0.00
	101	-1.31	46.95	-0.24	0.00	0.00	0.00
	102	-7.90	28.62	-0.10	0.00	0.00	0.00
	103	-15.01	49.81	-0.17	0.00	0.00	0.00
	104	-14.66	36.19	-0.10	0.00	0.00	0.00
	105	-0.91	16.84	-0.07	0.00	0.00	0.00
	106	-1.03	26.25	-0.13	0.00	0.00	0.00
	107	-0.68	12.63	-0.06	0.00	0.00	0.00
	201	-1.01	32.85	-0.17	0.00	0.00	0.00
	202	-9.74	47.58	-0.19	0.00	0.00	0.00
	203	-9.49	28.75	-0.09	0.00	0.00	0.00
	204	-1.01	32.85	-0.17	0.00	0.00	0.00
	205	-0.76	14.03	-0.06	0.00	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

```
120. ************************************************************
121. LOAD LIST 201 TO 205
122. PRINT JOINT DISPLACEMENTS LIST 5 TO 8
```

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
5	201	-0.0002	-0.0349	-0.0048	-0.0000	-0.0001	-0.0000
	202	0.0669	-0.0293	-0.0058	-0.0000	0.0004	0.0014
	203	0.0662	-0.0089	-0.0010	-0.0000	0.0004	0.0014
	204	-0.0002	-0.0349	-0.0048	-0.0000	-0.0001	-0.0000
	205	-0.0009	-0.0145	0.0000	-0.0000	-0.0001	-0.0000
6	201	-0.0002	-0.0466	-0.0241	-0.0000	0.0001	0.0000
	202	0.0669	-0.0522	-0.0231	-0.0000	0.0006	0.0015
	203	0.0662	-0.0219	-0.0041	-0.0000	0.0005	0.0014
	204	-0.0002	-0.0466	-0.0241	-0.0000	0.0001	0.0000
	205	-0.0009	-0.0163	-0.0051	-0.0000	0.0000	0.0000
7	201	0.0049	-0.0301	-0.0049	-0.0000	-0.0001	-0.0000
	202	0.0715	-0.0247	-0.0059	-0.0000	-0.0003	0.0014
	203	0.0671	-0.0079	-0.0010	-0.0000	-0.0003	0.0014
	204	0.0049	-0.0301	-0.0049	-0.0000	-0.0001	-0.0000
	205	0.0005	-0.0133	-0.0000	-0.0000	-0.0001	-0.0000
8	201	0.0048	-0.0343	-0.0239	-0.0001	0.0000	0.0008
	202	0.0716	-0.0396	-0.0230	-0.0001	0.0000	0.0009
	203	0.0672	-0.0186	-0.0041	-0.0000	0.0000	0.0003
	204	0.0048	-0.0343	-0.0239	-0.0001	0.0000	0.0008
	205	0.0005	-0.0133	-0.0051	-0.0000	0.0000	0.0003

********** END OF LATEST ANALYSIS RESULT **********

```
123. ************************************************************
124. LOAD LIST 100 TO 107
125. PARAMETER 1
126. CODE LRFD
127. FU 417940 ALL
128. FYLD 275000 ALL
129. BEAM 1 ALL
130. NSF 0.9 ALL
131. TRACK 0 ALL
132. LX 0.8 MEMB 27 29 30
133. LY 0.8 MEMB 27 29 30
134. LZ 6 MEMB 21 TO 26 43 TO 48
135. LZ 5.25 MEMB 10 14
136. LZ 5.35 MEMB 15 19
137. LZ 4.825 MEMB 3 11
138. LZ 2.4 MEMB 7 8 65 66 69 71 75 81
139. LZ 3 MEMB 5 6 41 42 67 68 73 74
140. LY 4 MEMB 72 82 86
```

```
141. LY 2.4 MEMB 72 82 86
142. *KY 0.85 MEMB 31 TO 40 49 TO 64
143. *KZ 0.85 MEMB 31 TO 40 49 TO 64
144. KY 0.85 MEMB 31 TO 37 39 49 TO 64
145. KZ 0.85 MEMB 31 TO 37 39 49 TO 64
146. CHECK CODE ALL
```

STAAD.Pro CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.103	103
		16.22 C	0.82	0.15	2.20
2 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.097	103
		13.38 C	0.82	0.04	2.20
3 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.134	103
		42.66 C	0.82	0.01	2.20
4 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.136	103
		38.82 C	0.81	0.37	2.20
5 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.023	101
		0.00 T	-0.01	1.24	1.00
6 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.006	101
		0.00 T	-0.00	0.29	1.00
7 ST	UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.011	100
		0.10 C	-0.00	-0.26	1.20
8 ST	UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.115	101
		0.52 T	-0.00	-2.81	1.20
9 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.102	103
		12.60 C	0.82	0.15	0.00
10 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.096	103
		9.95 C	0.82	0.04	0.00
11 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.134	103
		33.77 C	0.82	0.01	0.00
12 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.133	103
		28.71 C	0.81	0.37	0.00
13 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.125	103
		14.18 C	1.05	0.09	2.63
14 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.120	103
		12.61 C	1.05	-0.00	2.63
15 ST	IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.126	103
		16.14 C	1.04	0.01	2.63

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
16	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.106	103
		10.79 C	0.89	0.17	2.63
17	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.104	103
		1.16 C	1.05	0.09	0.00
18	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.102	103
		1.17 C	1.05	-0.00	0.00
19	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.119	103
		11.13 C	1.04	0.01	0.00
20	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.098	103
		5.73 C	0.89	0.17	0.00
21	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.100	103
		12.38 C	0.21	-0.00	3.00
22	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.081	103
		8.59 C	0.21	-0.00	3.00
23	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.056	103
		3.57 C	0.21	-0.00	3.00
24	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.098	103
		11.97 C	0.22	0.00	3.00
25	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.080	103
		8.23 C	0.21	0.00	3.00
26	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.048	104
		3.97 C	0.16	0.00	3.00
27	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.293	101
		0.65 T	0.00	-14.15	1.20
29	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.293	101
		0.51 T	0.00	-14.15	1.20
30	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.293	101
		0.29 T	0.00	-14.15	1.20
31	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.032	100
		0.01 T	0.00	-0.15	1.63

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
32	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.037	101
		1.13 C	0.00	-0.14	1.78
33	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.035	100
		0.00 T	0.00	-0.16	1.78
34	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.033	100
		0.28 C	0.00	-0.15	1.63
35	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.035	100
		0.07 C	0.00	-0.16	1.78
36	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.036	100
		0.25 C	0.00	-0.16	1.78
37	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.037	100
		0.20 C	0.00	-0.16	1.82
39	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.036	100
		0.11 T	0.00	-0.16	1.82
41	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.293	101
		0.51 C	0.01	-14.03	2.00
42	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.132	101
		0.05 C	0.00	-6.32	2.00
43	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.052	103
		13.74 T	0.21	0.00	0.00
44	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.049	103
		10.08 T	0.21	0.00	0.00
45	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.047	100
		2.01 T	0.25	0.00	0.00
46	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.052	103
		13.36 T	0.22	-0.00	0.00
47	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.048	103
		9.63 T	0.21	-0.00	0.00
48	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.048	103
		9.19 T	0.21	-0.00	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
49	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.061	103
		15.19 T	0.00	-0.18	1.86
50	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.288	103
		17.20 C	0.00	-0.18	1.86
51	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.058	103
		11.48 T	0.00	-0.19	1.99
52	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.264	103
		13.33 C	0.00	-0.19	1.99
53	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.056	100
		0.91 C	0.00	-0.22	1.99
54	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.116	103
		8.69 C	0.00	-0.19	1.99
55	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.081	101
		4.42 C	0.00	-0.19	2.03
56	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.093	103
		5.77 C	0.00	-0.19	2.03
57	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.060	103
		14.70 T	0.00	-0.18	1.86
58	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.281	103
		16.70 C	0.00	-0.18	1.86
59	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.059	100
		1.24 C	0.00	-0.22	1.99
60	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.255	103
		12.79 C	0.00	-0.19	1.99
61	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.059	100
		1.24 C	0.00	-0.22	1.99
62	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.112	103
		8.21 C	0.00	-0.19	1.99
63	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.131	101
		10.10 C	0.00	-0.19	2.03

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
64	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.119	101
		8.75 C	0.00	-0.19	2.03
65	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.011	100
		0.01 C	-0.00	-0.26	0.00
66	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.115	101
		0.11 T	-0.00	-2.81	0.00
67	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.052	101
		0.19 C	0.00	2.50	2.00
68	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.109	101
		0.48 T	-0.00	5.23	2.00
69	ST L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.110	100
		0.14 C	-0.00	-0.34	1.20
70	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.007	101
		0.61 C	0.00	-0.01	0.50
71	ST L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.110	100
		0.14 C	-0.00	-0.34	0.00
72	ST L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.101	100
		0.15 C	0.00	-0.31	1.00
73	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.257	101
		0.00 C	0.01	-14.03	0.00
74	ST IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.116	101
		0.20 C	0.00	-6.32	0.00
75	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.190	101
		0.60 T	-0.01	-10.53	1.10
76	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.014	100
		0.18 T	0.00	-0.02	0.78
77	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.014	100
		0.05 T	0.00	-0.02	0.78
78	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.017	101
		0.79 C	0.00	-0.02	0.78

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
79	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.014	100
		0.06 C	0.00	-0.02	0.78
80	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.007	101
		0.67 T	0.00	-0.01	0.50
81	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.181	101
		0.60 T	-0.01	-9.98	0.00
82	ST L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.120	100
		0.01 T	-0.00	-0.37	1.00
86	ST L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.101	100
		0.17 T	-0.00	-0.31	0.00
87	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.014	100
		0.12 T	0.00	-0.02	0.78
88	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.015	101
		0.51 C	0.00	-0.02	0.78
89	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.014	100
		0.12 T	0.00	-0.02	0.78
90	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.015	101
		0.51 C	0.00	-0.02	0.78

********** END OF TABULATED RESULT OF DESIGN **********

147. STEEL TAKE OFF ALL

STEEL TAKE-OFF

	PROFILE	**LENGTH(METE)**	**WEIGHT(KN)**
ST	IPE200	52.70	11.538
ST	UPN140	40.80	6.394
ST	UPN200	9.60	2.375
ST	L90X90X8	90.98	9.715
ST	L75X75X8	6.40	0.565
ST	L65X65X6	14.50	0.839
		TOTAL =	31.425

********** END OF DATA FROM INTERNAL STORAGE **********

148. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 14,2020 TIME= 9:39:38 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```

ATTACHMENT 3 F1 STAAD OUTPUT

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    SEP 17, 2020                *
*           Time=    15:59:39                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\MAADEN\Potable Water Treatment Plant\STAIR\STAAD\F1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 09-SEPT-20
     4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
     5. JOB CLIENT MA'ADEN
     6. JOB NO 49WF
     7. ENGINEER NAME BJB
     8. CHECKER NAME SJM
     9. APPROVED NAME AG
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT METER KN
    13. JOINT COORDINATES
    14. 1 1.2 0 0; 2 1.65 0 0; 3 4.05 0 0; 7 4.5 0 0
    15. MEMBER INCIDENCES
    16. 1 1 2; 2 2 3; 3 3 7
    17. DEFINE MATERIAL START
    18. ISOTROPIC CONCRETE
    19. E 2.17185E+007
    20. POISSON 0.17
    21. DENSITY 23.5616
    22. ALPHA 1E-005
    23. DAMP 0.05
    24. TYPE CONCRETE
    25. STRENGTH FCU 27579
    26. END DEFINE MATERIAL
    27. MEMBER PROPERTY AMERICAN
    28. 1 TO 3 PRIS YD 0.5 ZD 2.4
    29. CONSTANTS
    30. MATERIAL CONCRETE ALL
    31. SUPPORTS
    32. 2 3 FIXED BUT MY MZ KFY 0.2
    33. LOAD 100 LOADTYPE NONE  TITLE 1.4 ( D )
    34. MEMBER LOAD
    35. 1 TO 3 UNI GY -28.86
    36. 1 TRAP GY 114.49 113.53
    37. 2 TRAP GY 113.53 108.42
    38. 3 TRAP GY 108.42 107.46
```

```
39. JOINT LOAD
40. 2 FY -23.8 MZ 4.64
41. 3 FY -22.4
42. LOAD 101 LOADTYPE NONE  TITLE 1.2( D  ) + 1.6 L
43. MEMBER LOAD
44. 1 TO 3 UNI GY -28.86
45. 1 TRAP GY 119.96 119.14
46. 2 TRAP GY 119.14 114.76
47. 3 TRAP GY 114.76 113.94
48. JOINT LOAD
49. 2 FY -60.4 MZ 3.97
50. 3 FY -51.2
51. LOAD 102 LOADTYPE NONE  TITLE 1.2( D ) + 0.8 WLX
52. MEMBER LOAD
53. 1 TO 3 UNI GY -28.86
54. 1 TRAP GY 107.42 102.09
55. 2 TRAP GY 102.09 73.62
56. 3 TRAP GY 73.62 68.28
57. JOINT LOAD
58. 2 FY -8.4 MZ -25.83
59. 3 FY -7.2
60. LOAD 103 LOADTYPE NONE  TITLE 1.2( D  ) + 0.5 L + 1.6 WLX
61. MEMBER LOAD
62. 1 TO 3 UNI GY -28.86
63. 1 TRAP GY 129.55 118.05
64. 2 TRAP GY 118.05 56.74
65. 3 TRAP GY 56.74 45.24
66. JOINT LOAD
67. 2 FY -8.9 MZ -55.64
68. 3 FY -5.2
69. LOAD 104 LOADTYPE NONE  TITLE 0.9 ( D ) + 1.6 WLX
70. MEMBER LOAD
71. 1 TO 3 UNI GY -28.86
72. 1 TRAP GY 99.71 88.01
73. 2 TRAP GY 88.01 25.6
74. 3 TRAP GY 25.6 13.9
75. JOINT LOAD
76. 2 FY 8.7 MZ -56.64
77. 3 FY 9.6
78. LOAD 105 LOADTYPE NONE  TITLE 1.2( D ) + 0.8 WLZ
79. MEMBER LOAD
80. 1 TO 3 UNI GY -28.86
81. 1 TRAP GY 98.14 97.31
82. 2 TRAP GY 97.31 92.93
83. 3 TRAP GY 92.93 92.11
84. JOINT LOAD
85. 2 FY 54 MZ 3.97
86. 3 FY -93.6
87. LOAD 106 LOADTYPE NONE  TITLE 1.2( D  ) + 0.5 L + 1.6 WLZ
88. MEMBER LOAD
89. 1 TO 3 UNI GY -28.86
90. 1 TRAP GY 104.96 104.14
91. 2 TRAP GY 104.14 99.76
92. 3 TRAP GY 99.76 98.94
93. JOINT LOAD
94. 2 FY 115 MZ 3.97
```

```
  95. 3 FY -178
  96. LOAD 107 LOADTYPE NONE  TITLE 0.9 ( D ) + 1.6 WLZ
  97. MEMBER LOAD
  98. 1 TO 3 UNI GY -28.86
  99. 1 TRAP GY 73.61 72.99
 100. 2 TRAP GY 72.99 69.7
 101. 3 TRAP GY 69.7 69.08
 102. JOINT LOAD
 103. 2 FY 133.5 MZ 2.98
 104. 3 FY -163
 105. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**4**	**NUMBER OF MEMBERS**	**3**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 9 DOF
TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 18
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 324568.1 MB

```
**NOTE - NEAR INSTABILITY AT JOINT     7   DIRECTION = FY
 PROBABLE CAUSE  - NEAR SINGULAR K-MATRIX
 K-MATRIX DIAG=  9.2819066E+04 L-MATRIX DIAG=  2.2840548E-03 EQN NO      14

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


  106. PRINT MEMBER FORCES ALL
```

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
1	100	1	0.00	-0.00	0.00	0.00	0.00	0.00
		2	0.00	-38.32	0.00	0.00	0.00	8.64
	101	1	0.00	-0.00	0.00	0.00	0.00	-0.00
		2	0.00	-40.81	0.00	0.00	0.00	9.20
	102	1	0.00	0.00	0.00	0.00	0.00	-0.00
		2	0.00	-34.15	0.00	0.00	0.00	7.77
	103	1	0.00	0.00	0.00	0.00	0.00	0.00
		2	0.00	-42.72	0.00	0.00	0.00	9.81
	104	1	0.00	-0.00	0.00	0.00	0.00	0.00
		2	0.00	-29.25	0.00	0.00	0.00	6.78
	105	1	0.00	0.00	0.00	0.00	0.00	0.00
		2	0.00	-30.99	0.00	0.00	0.00	6.99
	106	1	0.00	-0.00	0.00	0.00	0.00	0.00
		2	0.00	-34.06	0.00	0.00	0.00	7.68
	107	1	0.00	0.00	0.00	0.00	0.00	0.00
		2	0.00	-20.00	0.00	0.00	0.00	4.51
2	100	2	0.00	-97.90	0.00	0.00	0.00	-4.00
		3	0.00	-99.18	0.00	0.00	0.00	7.99
	101	2	0.00	-105.16	0.00	0.00	0.00	-5.23
		3	0.00	-106.26	0.00	0.00	0.00	8.64
	102	2	0.00	-88.75	0.00	0.00	0.00	-33.60
		3	0.00	-52.84	0.00	0.00	0.00	4.17
	103	2	0.00	-108.92	0.00	0.00	0.00	-65.45
		3	0.00	-31.56	0.00	0.00	0.00	2.05
	104	2	0.00	-72.91	0.00	0.00	0.00	-63.42
		3	0.00	5.84	0.00	0.00	0.00	-1.12
	105	2	0.00	-78.97	0.00	0.00	0.00	-3.02
		3	0.00	-80.06	0.00	0.00	0.00	6.43
	106	2	0.00	-87.16	0.00	0.00	0.00	-3.71
		3	0.00	-88.26	0.00	0.00	0.00	7.12
	107	2	0.00	-50.57	0.00	0.00	0.00	-1.53
		3	0.00	-51.39	0.00	0.00	0.00	4.09
3	100	3	0.00	-35.59	0.00	0.00	0.00	-7.99
		7	0.00	-0.00	0.00	0.00	0.00	0.00
	101	3	0.00	-38.47	0.00	0.00	0.00	-8.64
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	102	3	0.00	-18.94	0.00	0.00	0.00	-4.17
		7	0.00	0.00	0.00	0.00	0.00	0.00
	103	3	0.00	-9.96	0.00	0.00	0.00	-2.05
		7	0.00	0.00	0.00	0.00	0.00	-0.00

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
	104	3	0.00	4.10	0.00	0.00	0.00	1.12
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	105	3	0.00	-28.65	0.00	0.00	0.00	-6.43
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	106	3	0.00	-31.72	0.00	0.00	0.00	-7.12
		7	0.00	-0.00	0.00	0.00	0.00	-0.00
	107	3	0.00	-18.24	0.00	0.00	0.00	-4.09
		7	0.00	-0.00	0.00	0.00	0.00	-0.00

********** END OF LATEST ANALYSIS RESULT **********

107. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 17,2020 TIME= 15:59:40 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```